Step 6: Draw a scale for the data on the x axis

Step 7: Locate the five numbers in the scale and draw a box around

Box plot



Points to Remember

- The range is the difference between the largest and smallest observations.
- The inter quartile range (IQR) is the difference between the upper and lower quartiles.
- The variance is the average of the squares of the values of $x-\overline{x}$
- The standard deviation (s.d) is the square root of the variance and has the same units as x
- If a population is approximately **symmetric** in a sample the mean and the median will have similar values. Typically their values will also be close to that of the mode of the population (if there is one!)
- A population that is not symmetric is said to be **skewed**. A distribution with a long 'tail' of high values is said to be **positively skewed**, in which case the mean is usually greater than the mode or the median. If it has a long tail of low values it is said to be **negatively skewed**, then the mean is likely to be the lowest of the three location measures of the distribution
- Box plots (Box-whisker diagrams): indicate the least and greatest values together with the quartiles and the median.

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